

LESSON 5-1 • SESSION 2 • PRACTICE • FORM B**Area of Rhombuses**

Learning target – I can find the area of a rhombus using the base and height or the two diagonals. • 6.G.A.1

REMEMBER

- A **rhombus** has 4 equal sides. It is a parallelogram, so **$A = b \times h$** works.
- With two **diagonals**: **$A = d1 \times d2 \div 2$** .
- Missing diagonal: double the area, then divide ($24 \times 2 = 48$, $48 \div 6 = 8$).

WORKED EXAMPLE

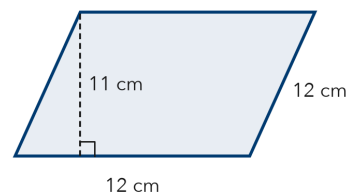
A rhombus has diagonals of 10 cm and 8 cm. Find its area.

Step 1 Use $A = d1 \times d2 \div 2$.**Step 2** $10 \times 8 = 80$ **Step 3** $80 \div 2 = 40$ **Answer** **40 cm²****WORDS** rhombus (*rombo*) • diagonal (*diagonal*) • congruent (*congruente*)**PRACTICE***show your work • label units*

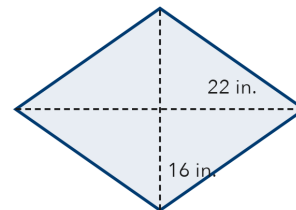
1. Find the area of each rhombus. Choose the formula that fits.

a. diagonals 12 m and 9 m _____**b.** base 13 in, height 8 in _____**c.** diagonals 16 m and 5 m _____**d.** base 7 cm, height 5 cm _____

- 2.
- Backsplash Tile.**
- A rhombus tile has sides of 12 cm and a height of 11 cm. Find its area.

Hint Any side can be the base.*Show your work.***Answer** _____ cm²

- 3.
- Paper Kite.**
- A kite is a rhombus. Find its area.

Hint Multiply the diagonals, then divide by 2.*Show your work.***Answer** _____ in²

- 4.
- Quilt Patch.**
- A rhombus patch has an area of 126 cm
- ²
- . Its sides are 14 cm. Find its height.

Answer _____ cm

- 5.
- Window.**
- A rhombus window has an area of 108 cm
- ²
- and a height of 9 cm. Find its base.

Answer _____ cm

- 6.
- Sail Panel.**
- A rhombus sail panel has diagonals of 20 ft and 26 ft. Find its area.

*Show your work.***Answer** _____ ft²

7. Missing Diagonal. A rhombus sign has an area of 45 m^2 . One diagonal is 9 m . Find the other diagonal.

Hint Double the area, then divide by 9.

Show your work.

Answer_____ m

8. Bike-Lane Marker. A rhombus marker has diagonals of 24 in. and 18 in. One strip of tape covers 54 in^2 .

Hint Doubling both diagonals multiplies the area by 2×2 .

- a. Find the area. _____ in^2
- b. How many strips cover it? _____
- c. Both diagonals are doubled. Find the new area. How many times larger is it? _____

Show your work.

FIND THE ERROR

explain the mistake, then fix it

9. Nia’s Work. A rhombus has diagonals of 12 ft and 9 ft . Nia wrote: “ $A = 12 \times 9 = 108\text{ ft}^2$.” Explain her mistake. Then find the area.

TEST PRACTICE

choose the best answer

- 10.** A rhombus has diagonals of 20 in. and 9 in. What is its area?
- A** 29 in^2 **B** 45 in^2 **C** 90 in^2 **D** 180 in^2
- 11.** Which expression finds the area of a rhombus from its diagonals?
- A** $d1 + d2$ **B** $d1 \times d2$ **C** $(d1 \times d2) \div 2$ **D** $2 \times d1 \times d2$

SELF-CHECK CODE

find your answer · copy its letter into that problem’s box · spell a math word

10 O	9 L	5 X	352 Q	520 Z	176 O	216 N	260 G	12 Y	132 P
-----------------------	----------------------	----------------------	------------------------	------------------------	------------------------	------------------------	------------------------	-----------------------	------------------------

#2	#3	#4	#5	#6	#7	#8a
----	----	----	----	----	----	-----

CHECK YOURSELF ☐ I can do this on my own ☐ I need a little help ☐ I have a question about #_____